Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 1 of 6 · Question 4 part a.i · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
4 (a)	(i)	• eat too much; • too little exercise; • increased fat storage;		3
	(ii)	• heart disease/attacks/strokes; • causes joints in legs to develop arthritis;	Allow high blood pressure/hypertension	2
(b)	(i)	C (pancreas); A is for excretion B is site of insulin action D produces other hormones		1
	(ii)	B (liver); A is for excretion C is site of production D produces other hormones		1
(c)	(i)	26378 - 22092 = 4286; $\frac{4286 \times 100}{22092}$; = 19.4%;	Allow $\frac{26378 - 22092}{22092}$ for 2 marks Ecf from first calculation max 2 marks	3
	(ii)	0.9 x 3.8 = 3.4(2) million;		1
(d)		any three from • less blood flows; • death of tissues; • increased risk of infection; • less white blood cells to deal with infection;		3

Total 14 marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 1 of 6 · Question 4 part a.i · 3 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 2 of 6 · Question 5 part b.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

5	(a)	(i)	175.9 + 178.5 + 166.3 + 167.6 ÷ 4;		1
		Clip with table	172.1;	Full marks for correct final answer Reject 172	1
		(ii)	Height increases with age (for males / females / both); Males are taller/females are shorter (at all ages); Females increase in height greater compared with males between the ages of 12 and 18; Greatest height increase between ages 12 to 14;		Max 2
		(iii)			1
			Use more students in each age group; Use a greater age range of students; Use ages with 1 year gap/13,15,17;	Allow other valid improvements Ignore repeats	1
	(b)	(i)	71 ÷ 1.46^2;		1
		Clip with table	33/33.3;	Full marks for final correct answer One mark for correct calculation with incorrect figures	1
		(ii)	student Y is obese;	ECF from b(i)	1
		CLIP (b)(i) with (b)(ii)	higher risk of diabetes/heart disease/arthritis/cancers;	Allow other medical condition linked to obesity	1

Total question 5 = 10 marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 2 of 6 · Question 5 part b.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 3 of 6 · Question 1 part b.ii · 1 mark

2020 JAN · Paper 01 · find part b.ii in this mark scheme extract

Number				
1	(a)	**The only correct answer is C** to slow down the growth of bacteria ***A** is incorrect because taste is personal preference* ***B** is incorrect because it does not kill the bacteria* ***D** is incorrect because the appearance is not affected*		1
	(b) (i)	protein		1
	(ii)	**The only correct answer is D** it does not contain energy from fat *A is incorrect because vitamins have no effect on body mass* *B is incorrect because cholesterol has no effect on body mass* *C is incorrect because it does contain energy from carbohydrates*		1
	(c)	• add Benedict's solution (to a sample of yogurt) (1) • heat gently (1) • brick red colour indicates the presence of sugar (1)	R if adding to glucose/reducing sugar Allow yellow, green, orange	3

Total for Question 1 = 6 marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 4 of 6 · Question 4 part a.ii · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
4 (a)	(i)	• eat too much; • too little exercise; • increased fat storage;		3
	(ii)	• heart disease/attacks/strokes; • causes joints in legs to develop arthritis;	Allow high blood pressure/hypertension	2
(b)	(i)	C (pancreas); A is for excretion B is site of insulin action D produces other hormones		1
	(ii)	B (liver); A is for excretion C is site of production D produces other hormones		1
(c)	(i)	26378 - 22092 = 4286; $\frac{4286 \times 100}{22092}$; = 19.4%;	Allow $\frac{26378 - 22092}{22092}$ for 2 marks Ecf from first calculation max 2 marks	3
	(ii)	0.9 x 3.8 = 3.4(2) million;		1
(d)		any three from • less blood flows; • death of tissues; • increased risk of infection; • less white blood cells to deal with infection;		3

Total 14 marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 4 of 6 · Question 4 part a.ii · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 5 of 6 · Question 3 part e · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

number			
3 (a)	C hypothalamus; A does not control sweating B does not control sweating D does not control sweating		1
(b)	B motor; A transmits impulse from sensory to motor C alternative name for A with same function D transmits impulse to brain		1
(c)	• (sweat) evaporates from (the surface of) the skin; • using heat from body/blood; • cools body/reduces body temperature;		3
(d)	• runs in families; • must be faulty/mutated gene/chromosome/allele; • which transfers/passes on information to offspring;		3
(e)	obesity / high sugar intake / high BMI;		1
(f)	• exercise/activity of individual varies; • greater rate of aerobic respiration; • heat given off as by-product/increased body temperature; • external temperature varies; • hotter more sweating/cooler less sweating; • mass of individual varies; • volume of water in the body/fluid intake varies;	Allow reverse argument for mps 2 and 3	5 Total 14

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 5 of 6 · Question 3 part e · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 6 of 6 · Question 3 part c.ii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c.ii in this mark scheme extract

Number				
3 (a)	(i)	3 320 000;		1
	(ii)	6.00239; x 10^5;	Allow 6.0024	2
	(iii)	pancreas;		1
(b)	(i)	120 ÷ 1.95^2/3.8029; 31.6;	Allow figure not rounded - 31.5581 Allow 31.56/32	2
	(ii)	obese;	ecf from previous question	1
(c)	(i)	• as BMI increases the risk of diabetes increases; • between a BMI of 24-35 the risk of diabetes increases more rapidly/BMI less than 25 the risk increases less rapidly;	Allow any range values between 24-35	2
	(ii)	Any two from the following: • balanced diet; • eat less fat; • eat less sugar	Ignore healthy diet Allow reduce alcohol/carbohydrate consumption for mp3	2

Total marks 11

Type MS · Explain causes of obesity, its health risks and dietary advice

Answer 6 of 6 · Question 3 part c.ii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks